HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Java Spring Boot** REST APIs for data product exposure, handling approximately 1,500 requests per minute and supporting seamless integration with AI agents.
- Developed Python data pipelines on **AWS Lambda** and **S3** to ingest and process enterprise data for AI model training, managing ~3,500 events/sec during peak hours.
- Contributed to the development of **Retrieval-Augmented Generation (RAG)** solutions, integrating enterprise data sources with LLM frameworks to enhance AI agent responses.
- Built APIs and automation frameworks using **Python** and **FastAPI** to expose data and business capabilities to AI agents and applications, improving data accessibility for downstream services.
- Containerized data platform services using **Docker** and deployed them to an **Azure Kubernetes Service (AKS)** cluster, configuring Helm charts for predictable rollouts and resource management.
- Collaborated with domain owners and data engineers to understand data products and build intelligent experiences that simplify data consumption through natural language interfaces.
- Participated in platform monitoring, troubleshooting, and continuous improvement activities, applying engineering best practices including automated testing and CI/CD with **GitHub Actions**.

SKILLS

AI/ML Platforms – AI Agents, Agentic Workflows, Model Context Protocol (MCP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases, Semantic Search

Backend Development – Java, Spring Boot, Python, FastAPI, Flask, REST APIs, gRPC, Event-Driven Architectures

Data Engineering – Apache Spark, Dremio, Azure Data Factory (ADF), Azure Databricks, Kafka, Data Pipelines, ETL

Cloud Platforms – Azure Data Platform Ecosystem, Azure Kubernetes Service (AKS), Azure Data Lake Storage (ADLS), Azure AI Services, AWS (EC2, S3, Lambda, CloudWatch)

DevOps & Tools – Docker, GitHub, GitHub Actions, Jira, Confluence, Dynatrace, Datadog

PROJECTS

AI Agent Integration for Data Discovery Platform

- Developed a **Python FastAPI** service to act as an intermediary between an LLM and enterprise data products, enabling natural language queries for data discovery.
- Implemented **Retrieval-Augmented Generation (RAG)** patterns to fetch relevant data context from a **Vector Database**, improving the accuracy of AI agent responses by approximately 30%.
- Designed and implemented API contracts for data product exposure, ensuring compatibility with agent frameworks and facilitating self-service data consumption for ~50 internal users.

Scalable Data Pipeline for AI Model Training

- Engineered a **Python** data pipeline using **AWS Lambda** and **S3** to ingest and transform large volumes of unstructured data, processing over 10,000 files daily.
- Integrated the pipeline with **Azure Databricks** for feature engineering and model training, contributing to a project that aimed to improve predictive accuracy for customer behaviour models.
- Wrote unit and integration tests for the pipeline components, achieving 85% code coverage and ensuring reliable data flow for downstream AI applications.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)